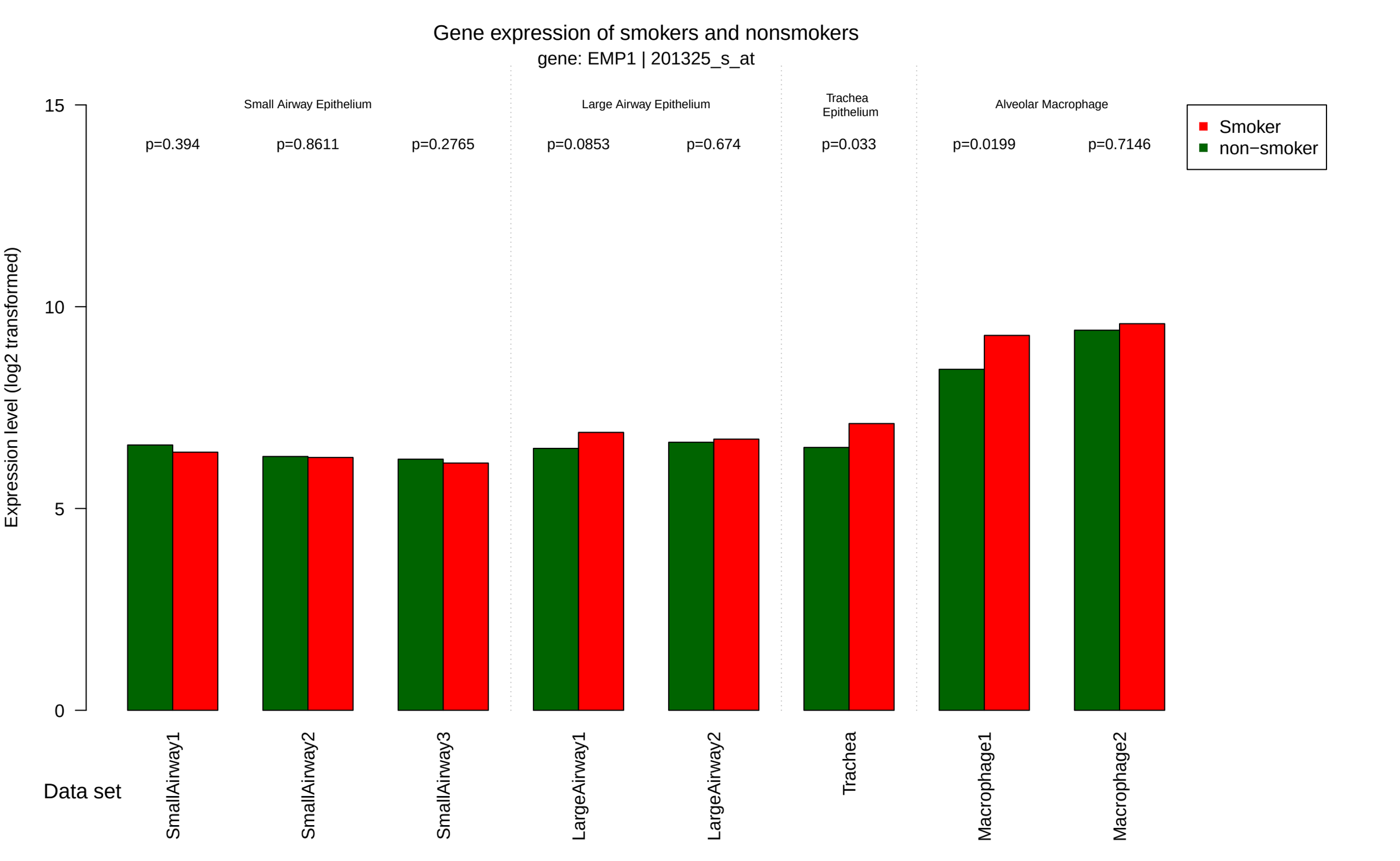
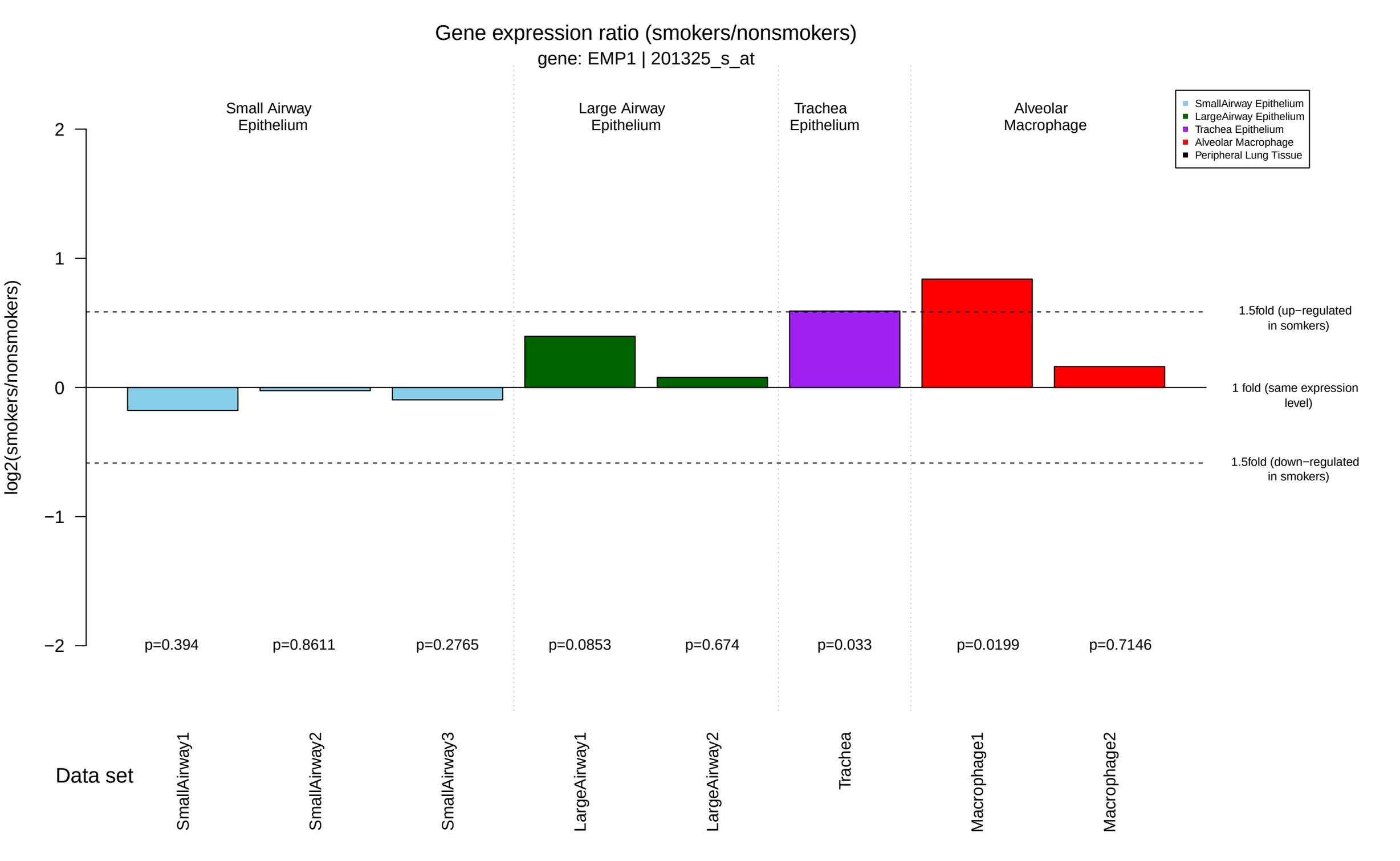
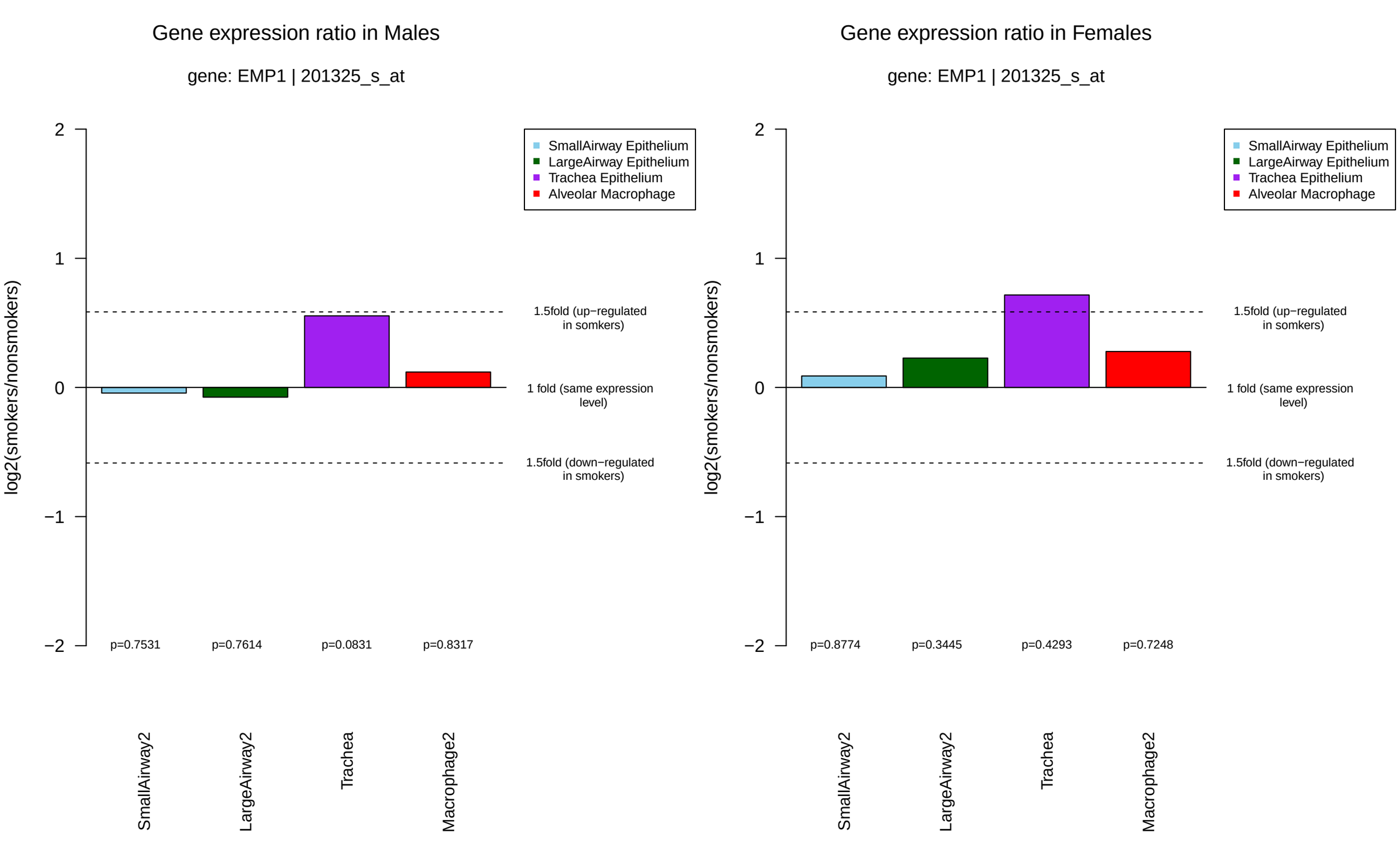
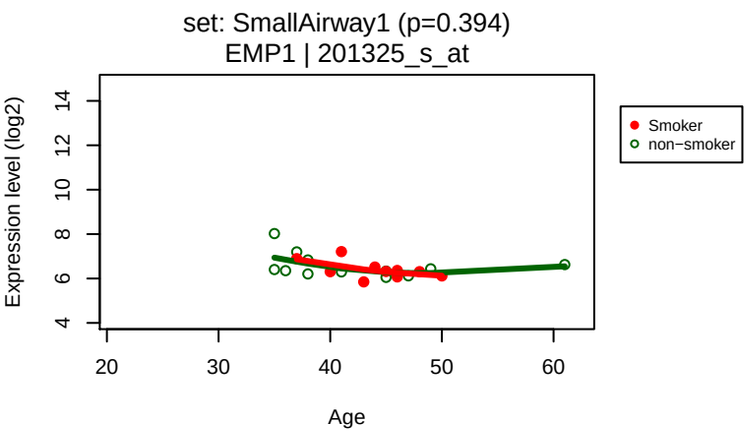
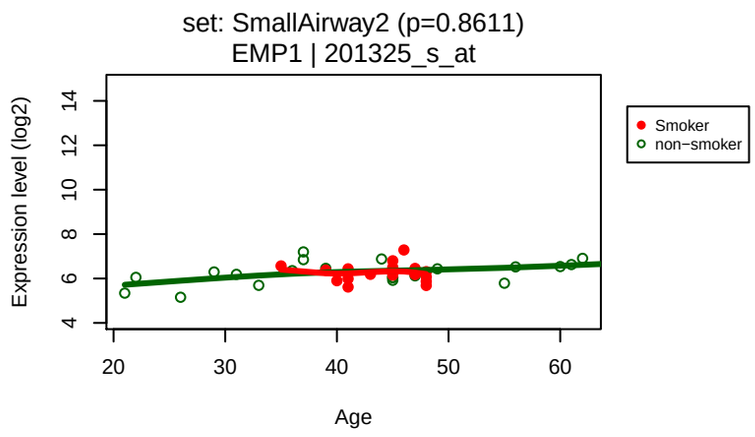
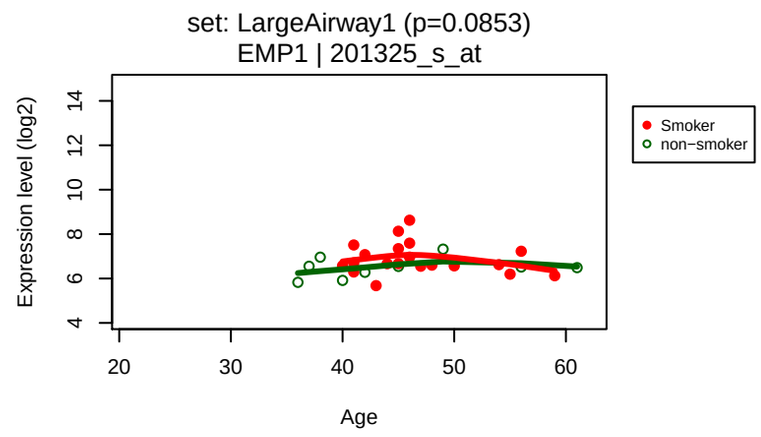

Gene expression of smokers and nonsmokers
gene: EMP1 | 201325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.394
p=0.8611
p=0.2765
p=0.0853
p=0.674
p=0.033
p=0.0199
p=0.7146
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

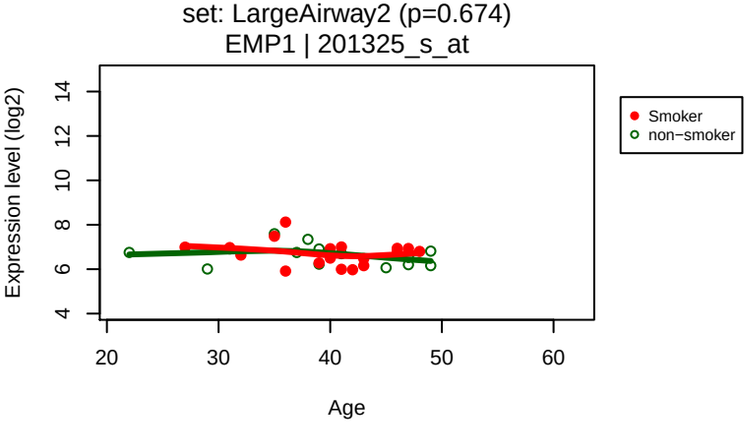

Gene expression ratio (smokers/nonsmokers)
gene: EMP1 | 201325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.394
p=0.8611
p=0.2765
p=0.0853
p=0.674
p=0.033
p=0.0199
p=0.7146
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

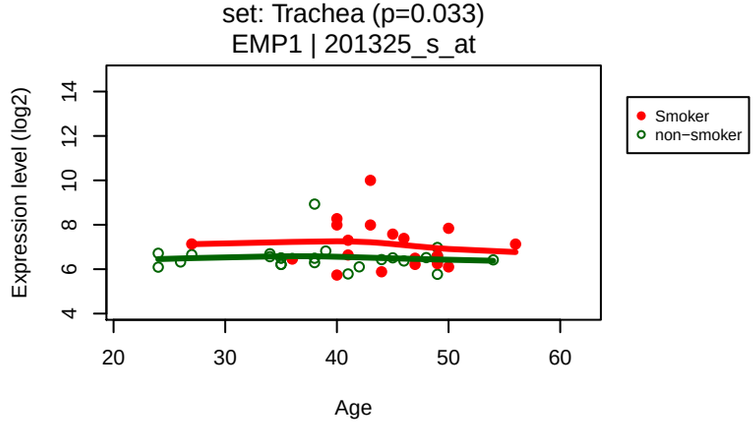

Gene expression ratio in Males
gene: EMP1 | 201325_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7531
p=0.7614
p=0.0831
p=0.8317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EMP1 | 201325_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8774
p=0.3445
p=0.4293
p=0.7248
SmallAirway2
LargeAirway2
Trachea
Macrophage2

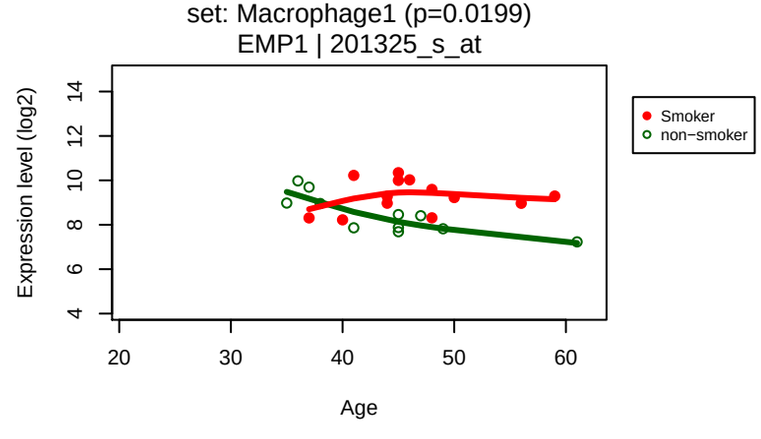

set: SmallAirway1 (p=0.394)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker

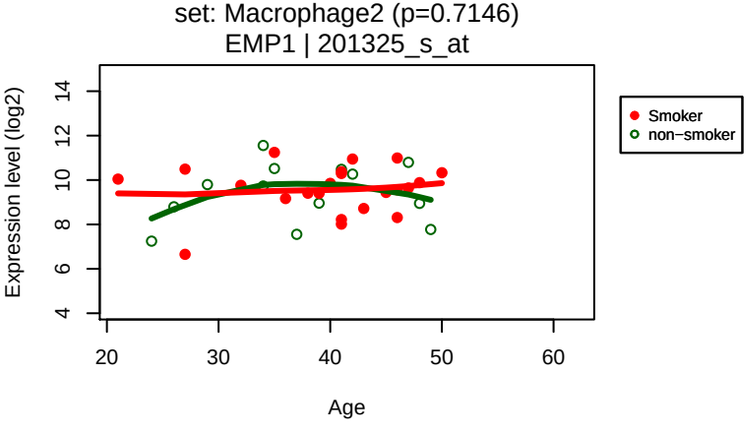

set: SmallAirway2 (p=0.8611)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker

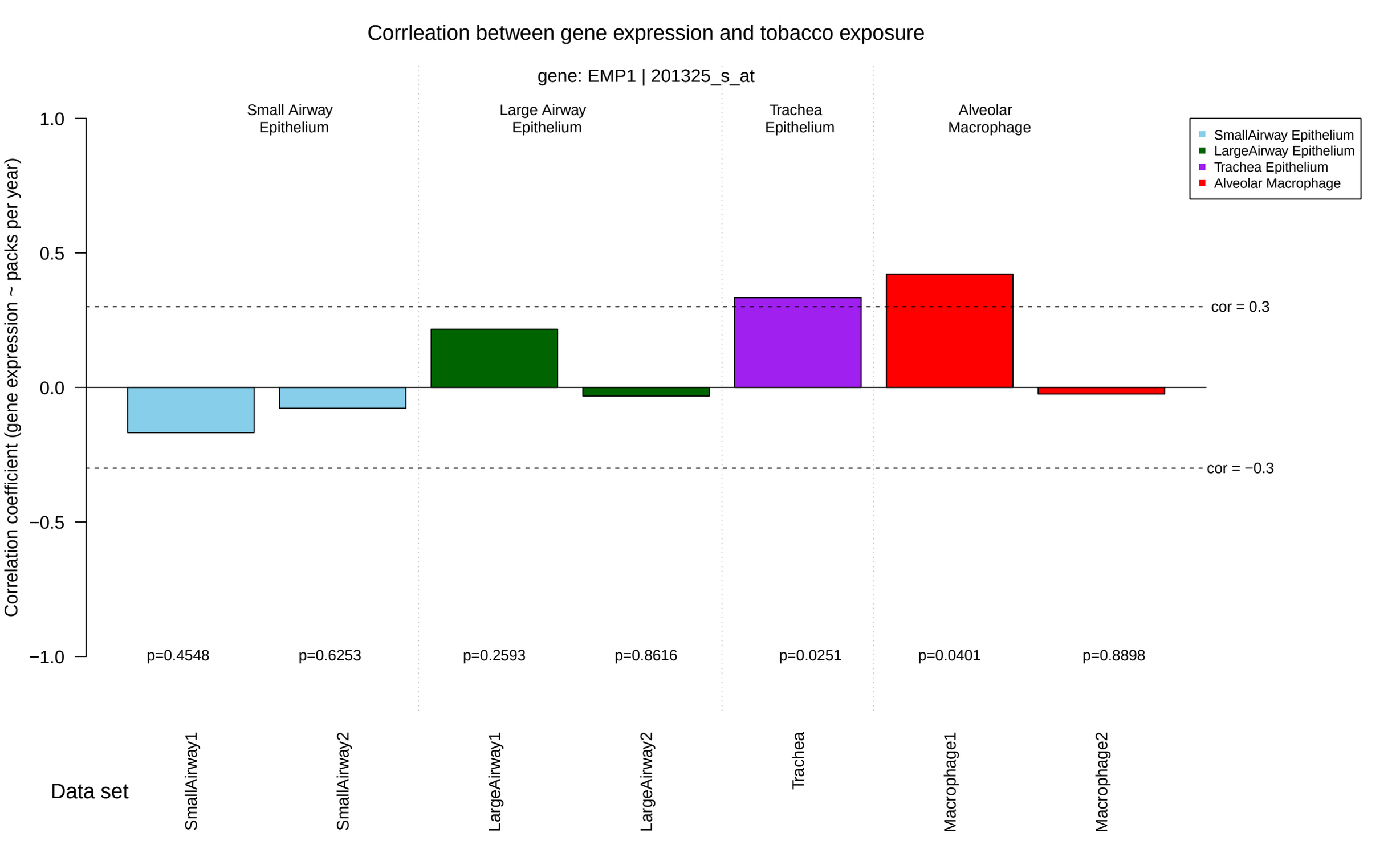

set: LargeAirway1 (p=0.0853)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.674)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.033)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0199)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7146)
EMP1 | 201325_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EMP1 | 201325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4548
p=0.6253
p=0.2593
p=0.8616
p=0.0251
p=0.0401
p=0.8898
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2